

LAYOUT_README

Bikera devboard – generated PCB layout (2026-07-05)

Generated from the anchor notes (00-04) + `Bikera PCB devboard.net`.

All 110 netlist components + 4 mounting holes placed, zero courtyard overlaps, every pad net-assigned and cross-checked against the netlist (0 mismatches).

Files

File	What
<code>Bikera PCB devboard.kicad_pcb</code>	The layout (also saved as <code>..._GENERATED_BACKUP.kicad_pcb</code>)
<code>Bikera PCB devboard.kicad_dru</code>	Custom DRC rules (RF width 0.85–0.95 mm, +BATT ≥0.5 mm, Power ≥0.35 mm)
<code>Bikera PCB devboard.kicad_pro</code>	Patched: 6 net classes + 24 net patterns (original in <code>.bak_before_netclasses</code>)
<code>layout_top_plan.png</code> / <code>layout_bottom_plan.png</code>	Placement plans with courtyards
<code>layout_pads_verification.png</code>	Pad-level render of the emitted file

KiCad was open while I wrote these (lock files present). If KiCad is still open: close it WITHOUT saving the PCB or project, then reopen. If it saved over anything, restore from `..._GENERATED_BACKUP.kicad_pcb` and `.bak_before_netclasses`.

⚠ Board size: 60×40 mm (not 50×30)

50×30 cannot hold this BOM: through-hole parts alone (SD header, ST-LINK area, DIP-8 socket, JSTs, buttons, 470 µF radial, USB-C...) block ~1100 mm² on *both* sides; with SMD courtyards the top side needed ~125% of a 50×30 board. You approved 60×40 (~78%/72% utilization). Anchor zones were kept, scaled: MCU center-left, GPS+antenna top-left, LoRa (bottom side) center-left, power column right edge, connectors on edges. Mounting holes M3 at (3.5, 3.5) from each corner. KiCad coords: origin top-left, Y down.

Placement highlights

- **U3 MCU** top, center (24, 26.2); C14/C15/C16 at VDD pins, C41 bulk adjacent, VDDA group (FB1, C17, C18, C40) at pin 9, all within anchor distances.
- **U9 RAK11720** on the **bottom side** (18.1, 24.5) — it's UART-only in the real netlist (not SPI as the anchors assumed), so routing is easy; C37–C39 + R24/R25 beside it.
- **U6 GPS** top-left, RF pin facing **AE1 (U.FL)** in the top-left corner; feed is ~5 mm. R28/C50/C49/D3 sit in the feed path. V_BCKP backup cluster (D2/D4/C32/U8) on the bottom under the GPS area.
- **Power column** on the right edge, top→bottom: L3, U4 (boost), U2 (GPS LDO), L1, IC2 (charger, BATT pads toward battery), L2, U1 (buck-boost). BT1 battery JST exits the right edge; TH1 next to it. **Star point**: bottom-right of the column, near (53, 33) — R16 servo shunt is right there.
- **J3 ST-LINK** is on the **bottom side** (debug adapter mates from below). **H3 SD header** lies along the bottom edge; **J7 ePaper** along the top edge; **H4 servo** bottom-right; **H2 accel header** vertical, left side.
- **IC1/IC3 hall sensors** are at the left/right board edges as placeholders — position them to match your magnet mechanics before routing.

⚠ Schematic bugs found in the netlist (fix in Eeschema)

1. **H3 pin 12** is in both `SPI1_MOSI` (pinfunction MOSI_12) and `SPI1_SCK` (SCK_12). I assigned it to **SPI1_MOSI** (pairs 11/12 = MOSI). Fix the SD adapter symbol/wiring.
2. **J7 pin 3** is in both `EPAPER_RST` and `SPI1_MOSI` (DIN_3). I assigned it to **SPI1_MOSI**; `EPAPER_RST` currently reaches no J7 pin on the board — it should probably be J7 pin 7 (pin 7 is absent from the netlist entirely).
3. **U9.12 (LORA_RF)** and **U9.33 (BLE_RF)** are **unconnected** in the netlist — the anchors describe a 50 Ω LoRa feed that doesn't exist in the schematic. If the RAK11720 needs a board-level antenna feed, add it and re-import.

RF keep-out — deliberate deviation from anchor 04

The anchors demand *no GND plane* in an 8 mm radius around the GPS antenna connector, but also a 0.9 mm/50 Ω microstrip — which *requires* the L2 GND reference underneath. Since AE1 is a U.FL to an **off-board** antenna (no on-board radiator), I kept **L2 (In1.Cu) solid** under the feed and applied the 8 mm copper-pour keep-out to **In2.Cu (power plane) and B.Cu** only (`GPS_ANT_KEEPOUT` zone). If you want the full anchor behaviour, edit the rule area and add In1.Cu — but then re-calculate the feed width (it will no longer be 50 Ω against L2).

Zones (drawn, not yet filled)

- `GND_PLANE_L2` on In1.Cu (whole board), `PWR_PLANE_L3` (+3V3) on In2.Cu, `GND_POUR_B` on B.Cu. Press **B** in pcbnew to fill.
- +BATT / +SYS / +5V / +3V3_GPS are *not* on the plane – route them as ≥ 0.5 mm traces (the .kicad_dru enforces widths).

Suggested next steps

1. Close KiCad (if open), reopen the project. The board opens as a v6 file and KiCad 10 will upgrade it on first save – that's expected.
2. Schematic → fix the three netlist bugs above → **Update PCB from Schematic (F8)** with "Re-link footprints" off. Expect it to be clean apart from the H3/J7 fixes, the 4 mounting holes (board-only), and possibly the J1 pad-name mapping (netlist `A1_B12` vs footprint pads `A1` + `B12` – KiCad handles this the same way I did).
3. Fill zones, then route in anchor priority order: +BATT→IC2→U1→star point, GPS RF feed (0.9 mm, no vias), SPI (SD/ePaper), I^C, UART, GPIO.
4. Run DRC (the .kicad_dru rules load automatically).
5. A few non-critical resistors sit 10–17 mm from their ICs (R5–R9 charger status/TS network, R13, R26) – scoot them closer if you re-arrange anything.